

Bladder and Urinary Function

Bladder cancer and metabolism of carcinogens (NAT2), urinary symptoms, incontinence and enuresis, infections and other urinary diseases, calculations and pharmacogenetics of bladder drugs

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Unfavourable. Bladder cancer has a genetic predisposition above average in loci-based scores: percentiles 94.9 (15 variants), 86.1 (24 variants) and 86.9 (510 thousand variants). The genomic score of 1 million variants is typical (percentile 57.2) and the load of GWAS loci in 71.7. Sérgio F is a slow acetylator of NAT2 (NAT2*5/*5; marker rs1495741 A/A) and homozygous for the risk allele in **PSCA**, **8q24**, **APOBEC3**, **CCNE1** and **UGT1A**.

Favourable. Urinary function has little predisposition: urinary incontinence is in percentile 10.3 and other urinary diseases (including infections) in 21.8. The loci loads of incontinence, retention and enuresis (4.3) and urinary infections (4.0) are low. Urinary calculations are in percentile 78.0 of the score and in the 27.0 of the loci. CYP2D6 does not have the variants *4, *10 or *41.

What does it mean? The absolute risk of bladder cancer is low without tobacco, and tobacco is the factor that most multiplies the effect of slow NAT2. What this profile asks is to avoid aromatic amines and then investigate any blood in the urine.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 23 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 21 of the chip, 2 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 0 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 7 PGS Catalog scores calculated locally with 3.305.926 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVÁVEL/ALTO; ≤ 20 : MENOS PROVÁVEL/BAIXO; between 20 and 80: TÍPICO.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 183 loci, 158 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all scores, a higher percentile = more risk.

Bladder cancer

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Bladder cancer (genomic)	PGS004687 (Hu 2024, LDpred2, 1.08 million variants)	1.051.651 1.077.775 (weighted coverage 97%)	57,2	TIPIC	typical
Bladder cancer	PGS002017 (Privé 2022, LDpred2, 510 thousand variants)	500.447 510.453 (weighted coverage 97%)	86,9	MAIS PROVABLE	unfavourable
Bladder cancer	PGS012553 (Kothers 2023, 24 variants)	23 24 (weighted coverage 95%)	86,1	MAIS PROVABLE	unfavourable
Bladder cancer	PGS000782 (Kachuri 2020, 15 variants)	14 15 (weighted coverage 94%)	94,9	MAIS PROVABLE	unfavourable

Urinary function

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Urinary frequency and irritability of the bladder	PGS001387 (Tanigawa 2022, 333 variants)	278 333 (weighted coverage 83%)	10,3	MENOS PROVABLE	favourable
Other urinary tract diseases (including urinary infections)	PGS004494 (Jung 2024, LDpred2, 1.06 million variants)	1.036.079 1.059.939 (weighted coverage 97%)	21,8	TIPIC	typical
Urinary calculations	PGS002075 (Privé 2022, LDpred2, 731 thousand variants)	717.434 731.196 (weighted coverage 97%)	78,0	TIPIC	typical

4. Key variants by topic

Variants with described effect. The judgment refers to the risk of bladder cancer and urinary diseases; in pharmacogenetics it refers to exposure to drugs.

Bladder cancer: GWAS loci

Common loci of bladder cancer GWAS, each with a small effect (OR 1.1–1.25). Several act on the urothelium itself (PSCA, TP63, SLC14A1) or how cells accumulate mutations (APOBEC3, TERT).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PSCA	rs2294008	T/T	chip		T (2 copy/s)	unfavourable	Prostate stem cells antigen, very expressed in the urothelium: T allele is one of the strongest bladder cancer loci (OR ~1.15–1.2 by allele)	Strong
8q24 (CASC11/MYC)	rs9642880	T/T	chip		T (2 copy/s)	unfavourable	8q24 region near MYC: bladder cancer locus (OR ~1.2) [GWAS Catalog: Bladder cancer, T allele, OR 1.24, p=4E-38, PMID 24163127]	Strong
TACC3/FGFR3	rs798766	C/C	chip		T (0 copy/s)	favourable	4p16 (TACC3, neighbor of FGFR3, the most mutated gene in bladder tumors) [GWAS Catalog: Bladder cancer, T allele, OR 1.22, p=7E-25, PMID 24163127]	Strong
TP63	rs710521	T/C	chip		T (1 copy/s)	unfavourable (light)	TP63 (differentiation of the urothelium) [GWAS Catalog: Bladder cancer, T allele, OR 1.15, p=5E-14, PMID 37210288]	Moderate
TERT-CLPTM1L	rs401681	C/T	chip		C (1 copy/s)	unfavourable (light)	5p15.33 (telomerase) [GWAS Catalog: Bladder cancer, C allele, OR 1.12, p=4E-11, PMID 24163127]	Moderate
APOBEC3A/B	rs1014971	T/T	chip		T (2 copy/s)	unfavourable	22q13 (APOBEC3: enzymes that leave the typical mutational signature of bladder tumors) [GWAS Catalog: Bladder cancer, T allele, OR 1.13, p=1E-11, PMID 24163127]	Moderate
CCNE1	rs8102137	C/C	chip		C (2 copy/s)	unfavourable	Cyclin E1 (cell cycle) [GWAS Catalog: Bladder cancer, C allele, OR 1.13, p=1E-11, PMID 24163127]	Moderate
UGT1A	rs11892031	A/A	chip		A (2 copy/s)	unfavourable	UGT1A (glucuronidation of aromatic amines and other carcinogens) [GWAS Catalog: Bladder cancer, allele A, OR 1.23, p=1E-11, PMID 37210288]	Moderate
SLC14A1	rs10853535	C/T	chip		C (1 copy/s)	unfavourable (light)	Urea transporter (SLC14A1, Kidd blood group) [GWAS Catalog: Bladder cancer, C allele, OR 1.14, p=8E-16, PMID 37210288]	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CBX6/ APOBEC3A	rs5750711		T/T	imputation DR2=0.98	T (2 copy/s)	unfavourable	APOBEC3/CBX6 region [GWAS Catalog: Cancer of bladder (PheCode 189.2), allele T, beta 0.135 [0.11-0.16] unit increase, p=2E-20, PMID39024449]	Moderate

Metabolism of urinary carcinogens (aromatic amines)

Aromatic amines (tobacco smoke, very carbonized meat, some paints and industrial solvents) are carcinogens that concentrate in urine. NAT2 inactivates them by acetylation in the liver. Who is slow acetylator makes more active amines reach the bladder, and the extra risk is higher with exposure to tobacco. The deletions of GSTM1 and GSTT1 also count, but do not read on the chip.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NAT2	rs1801280	*5	C/C	chip	C (2 copy/s)	unfavourable	NAT2*5 (341T>C): slow acetylation of aromatic amines (smoker, paints, charred meat)	Strong
NAT2	rs1799930	*6	G/G	chip	A (0 copy/s)	favourable	NAT2*6 (590G>A): slow acetylation	Strong
NAT2	rs1799931	*7	G/G	chip	A (0 copy/s)	favourable	NAT2*7 (857G>A): slow acetylation	Strong
NAT2	rs1801279	*14	G/G	chip	A (0 copy/s)	favourable	NAT2*14 (191G>A): slow acetylation (rare)	Moderate
NAT2 (tag)	rs1495741	tag	A/A	chip	A (2 copy/s)	unfavourable	Marker of the acetylator phenotype (A/ A = slow); the slow acetylator has more bladder cancer, especially with exposure to tobacco smoke (OR ~1.3-1.5)	Strong
NQO1	rs1800566	P187S	G/G	chip	A (0 copy/s)	favourable	P187S (NQO1*2): less quinone reductase activity (benzene, quinones); weak association with bladder cancer	Weak
GSTP1	rs1695	I105V	A/G	chip	G (1 copy/s)	unfavourable (light)	I105V: lower conjugation of some carcinogens; weak association	Weak

Urinary function, incontinence and infections

Urinary symptoms, incontinence and urinary infections have little known genetic basis. The few existing GWAS loci have small effects.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
MCHR2	rs9376454		C/C	imputation DR2=1.00	T (0 copy/s)	favourable	Night enuresis locus (MCHR2, melanin-concentrating hormone receptor) [GWAS Catalog: Nocturnal enuresis, allele T, OR 1.18, p=1E-13, PMID 33453761]	Weak
TLR4	rs4986790	D299G	A/A	chip	G (0 copy/s)	without judgment	D299G: lower signaling of the receptor that recognizes uropathogenic E. coli; association with urinary infections (mixed results)	Weak

Pharmacogenetics of bladder drugs

The antimuscarinics of the hyperactive bladder and mirabegron depend on CYP2D6 and CYP3A. The tiroprisium is eliminated by the kidney, almost without hepatic metabolism, and passes little the blood-brain barrier.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CYP2D6	rs3892097	*4	C/C	chip	T (0 copy/s)	favourable	CYP2D6*4 (loss of function): metabolizing tolterodine, fesoterodine (in part), darifenacin and duloxetine	Strong
CYP2D6	rs1065852	*10	G/G	chip	A (0 copy/s)	favourable	CYP2D6*10 (reduced function)	Moderate
CYP2D6	rs28371725	*41	C/C	chip	T (0 copy/s)	favourable	CYP2D6*41 (reduced function)	Moderate
CYP3A5	rs776746	*3	C/T	chip	C (1 copy/s)	without judgment	CYP3A5*3 (non-expressor): solifenacin, darifenacin, mirabegron and oxybutynin pass through CYP3A	Strong

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each disease, with each allele oriented towards the risk, compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Bladder cancer	55 (53)	55 de 106	53	71,7	typical
Incontinence, retention and enuresis	11 (9)	2 de 18	8	4,3	favourable
Urinary infections and cystitis	12 (7)	3 de 14	7	4,0	favourable
Urinary calculations	105 (89)	65 de 178	89	27,0	typical

6. Favourable

- **Unprobable urinary symptoms:** urinary incontinence and irritability of the bladder in percentile 10.3; incontinence, retention and enuresis with loci load in 4.3 MCHR2 C/C without the enuresis allele. **Moderate.**
- **Infections and other urinary diseases:** score in percentile 21.8; load of urinary infections and cystitis in 4.0. TLR4 D299G absent (A/A). **Moderate / weak.**
- **Bladder cancer, non-risk loci:** TACC3/FGFR3 C/C; NAT2*6, *7 and *14 absent; NQO1 P187S absent. **Moderate.**
- **Genomic score of typical bladder cancer** (percentile 57.2, 1.05 million variants used). **Moderate.**
- **CYP2D6 without *4, *10 or *41:** probably normal metabolism of tolterodine, darifenacin and duloxetine (the chip does not read deletions or duplications of the gene). **Moderate.**

7. Unfavourable

- **NAT2 slow acetylator** (*5/*5: rs1801280 C/C; marker rs1495741 A/A): the most studied phenotype of bladder cancer (OR ~1.3–1.5, higher in smokers). **Strong.**
- **Scores based on high bladder cancer loci:** percentiles 94.9, 86.1 and 86.9; loci load in 71.7. Moderate (the genomic score, which is the most complete, is typical).
- **Homozygosity for risk alleles:** PSCA T/T, 8q24 T/T, APOBEC3 T/T and CBX6/APOBEC3A T/T, CCNE1 C/C and UGT1A A/A. 1 allele: TP63, TERT-CLPTM1L, SLC14A1 and GSTP1 I105V. **Moderate.**
- **Urinary calculations:** score in percentile 78.0 (typical to slightly high), with loci load in 27.0. **Weak.**
- **No judgment:** CYP3A5 *1/*3 (partial expressor; solifenacin, darifenacin, mirabegron and rapamycin).

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
DNA Damage and DNA Repair Deficit	Slow NAT2: more aromatic amines in the urothelium	Unfavourable
Somatic Mutation Accumulation	Bladder cancer: loci scores in the 94.9–86.9 percentiles; APOBEC3 in homozygosity (mutational signature APOBEC)	Unfavourable (moderate)
Genomic instability	APOBEC3A/B and CBX6 with the risk allele in homozygosity	Unfavourable (mild)
Hayflick's Telomeres and Limit	TERT-CLPTM1L rs401681 with 1 risk allele	Neutral
Chronic Inflammation	Infections and other urinary diseases in percentile 21.8; loci of infections in 4.0	Favourable
Dysbiosis	Urinary microbiome: no genetic predisposition for infections; dapagliflozin increases urine glucose	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Blood in urine is always investigated:** with slow NAT2 and high loci scores, an episode of red or blood urine in a routine urine strip justifies urological evaluation, even if it is an episode alone and without pain. There is no screening for bladder cancer recommended for those who do not have symptoms. *Suggestion (not applied).*
2. **Avoid aromatic amines:** tobacco in any form and passive smoke; very carbonized or smoked meat (the plan already avoids it); professional or domestic exposure to paints, solvents, rubber and dyes; dark permanent hair paints. With slow NAT2, these are the exposures that count. *Suggestion (not applied).*
3. **Regular hydration and daily crucifers:** urination more often reduces the time of contact of the urothelium with the carcinogens. Crucifer isothiocyanates (broccoli, broccoli shoots, kale) are associated with less bladder cancer. High fiber (the more the better). *Suggestion (not applied).*
4. **Water:** if the water is well or hole, confirm that it does not have arsenic, a carcinogen of the bladder. *Suggestion (not applied).*
5. **Dapagliflozin:** The genetic predisposition to urinary infections is low, but the drug increases genital and urinary infections. Hygiene after urination and hydration; burning to urinate, fever or low back pain justify rapid observation. *Suggestion (not applied).*
6. **If one day a drug is needed for the hyperactive bladder:** prefer mirabegron or vibegron, or the trospium among antimuscarins. Antimuscarines that pass the blood-brain barrier (oxybutynin, tolterodine) add anticholinergic load, associated with cognitive decline. The tadalafil 5 mg of the plan already has an effect on low urinary symptoms. *Suggestion (not applied).*
7. **Calculations:** with the typical to slightly high score, maintaining a diuresis of at least 2-2.5 L/day, moderate salt and citrate of food (lime, orange). The details are in the kidney report. *Suggestion (not applied).*

Not recommended: pioglitazone (signal of bladder cancer in some studies), if it is ever considered for glucose; tobacco, cigarette or electronic cigarette; supplements with aristolochic acid (some plants from traditional Chinese medicine).

10. Limitations

- A genotyping chip with imputation does not read the deletions of GSTM1 and GSTT1 (zero GSTM1 is an established risk factor for bladder cancer), nor the deletions and duplications of CYP2D6.
- The 15 and 24 variants use only the strongest loci. The genomic score of 1 million variants is the most complete and is typical. The truth is probably between the two.
- The absolute risk of bladder cancer depends mainly on tobacco and professional exposures. A high score does not mean disease.
- Urinary symptoms, incontinence and infections have little known genetic basis and are greatly influenced by prostate, drugs and habits.
- The load of loci GWAS sums SNPs in linkage imbalance. It is a descriptive complement and does not replace the polygenic scores.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The alleles of each location are read from the Imputed VCF. The published effect allele is compared with those alleles and, if necessary, converted to the complementary strand (only in non-palindromic SNPs). In palindromics, strand + (GWAS Catalog standard and the harmonized files of PGS Catalog) is assumed. In the loci of GWAS Catalog with protective effect, the allele of effect is exchanged by the risk allele.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog: PGS004687 (Hu 2024), PGS002017 and PGS002075 (Privé 2022), PGS012553 (Kothers 2023), PGS000782 (Kachuri 2020), PGS001387 (Tanigawa 2022) and PGS004494 (Jung 2024). Score = Δ coefficient $\times$ dose of effect allele. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog (associations with $p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** GWAS Catalog, ClinVar, dbSNP/Ensembl and reference literature: Rothman 2010 and Figueroa 2014 (bladder cancer locus), García-Closas 2005 and 2011 (NAT2 and GSTM1), Koutros 2023 (GWAS), GWAS of clinical phenotypes of 2024 (incontinence and retention) and CPIC guidelines for CYP2D6. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Bladder cancer: GWAS loci	PSCA	rs2294008	T/T	chip	T	2	strand +	2 risk allele copies
Bladder cancer: GWAS loci	8q24 (CASC11/ MYC)	rs9642880	T/T	chip	T	2	strand +	2 risk allele copies
Bladder cancer: GWAS loci	TACC3/FGFR3	rs798766	C/C	chip	T	0	strand +	without the risk allele
Bladder cancer: GWAS loci	TP63	rs710521	T/C	chip	T	1	strand +	1 copy of risk allele
Bladder cancer: GWAS loci	TERT-CLPTM1L	rs401681	C/T	chip	C	1	strand +	1 copy of risk allele
Bladder cancer: GWAS loci	APOBEC3A/B	rs1014971	T/T	chip	T	2	strand +	2 risk allele copies
Bladder cancer: GWAS loci	CCNE1	rs8102137	C/C	chip	C	2	strand +	2 risk allele copies
Bladder cancer: GWAS loci	UGT1A	rs11892031	A/A	chip	A	2	strand +	2 risk allele copies
Bladder cancer: GWAS loci	SLC14A1	rs10853535	C/T	chip	C	1	strand +	1 copy of risk allele
Bladder cancer: GWAS loci	CBX6/ APOBEC3A	rs5750711	T/T	imputation DR2=0.98	T	2	strand +	2 risk allele copies
Metabolism of urinary carcinogens (aromatic amines)	NAT2	rs1801280	C/C	chip	C	2	strand +	2 risk allele copies
Metabolism of urinary carcinogens (aromatic amines)	NAT2	rs1799930	G/G	chip	A	0	strand +	without the risk allele
Metabolism of urinary carcinogens (aromatic amines)	NAT2	rs1799931	G/G	chip	A	0	strand +	without the risk allele
Metabolism of urinary carcinogens (aromatic amines)	NAT2	rs1801279	G/G	chip	A	0	strand +	without the risk allele
Metabolism of urinary carcinogens (aromatic amines)	NAT2 (tag)	rs1495741	A/A	chip	A	2	strand +	2 risk allele copies
Metabolism of urinary carcinogens (aromatic amines)	NQO1	rs1800566	G/G	chip	A	0	strand +	without the risk allele
Metabolism of urinary carcinogens (aromatic amines)	GSTP1	rs1695	A/G	chip	G	1	strand +	1 copy of risk allele
Urinary function, incontinence and infections	MCHR2	rs9376454	C/C	imputation DR2=1.00	T	0	strand +	without the risk allele
Urinary function, incontinence and infections	TLR4	rs4986790	A/A	chip	G	0	strand +	0 copy of the effect allele
Pharmacogenetics of bladder drugs	CYP2D6	rs3892097	C/C	chip	T	0	strand +	without the risk allele
Pharmacogenetics of bladder drugs	CYP2D6	rs1065852	G/G	chip	A	0	strand +	without the risk allele
Pharmacogenetics of bladder drugs	CYP2D6	rs28371725	C/C	chip	T	0	strand +	without the risk allele
Pharmacogenetics of bladder drugs	CYP3A5	rs776746	C/T	chip	C	1	strand +	1 copy of effect allele